When Auxiliary Losses Fail: Non-Stationary Targets Induce Directional Gradient Noise

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Abstract

Auxiliary losses often stabilize representation learning, yet sometimes degrade late-training performance in ways that look like ordinary RL noise. We identify a learning-dynamics mechanism governing this failure: when auxiliary targets are both *structured* (coupled to the current task state) and *non-stationary* (drifting as other learners update), they inject directional gradient noise that dominates policy gradients near convergence. The effect is not magnitude imbalance, not multi-task conflict, and not capacity consumption; it is driven by target drift across learners. We test this principle in cooperative MARL where co-learning teammates provide the natural non-stationarity. A target-source distinguishing experiment holds aux capacity, architecture, and the aux-to-encoder gradient pathway fixed and varies only the target: co-learning teammate targets are pathological (463 ± 9, $d = +1.40$ vs No Aux with bootstrap CI above zero); frozen-initial-policy targets (structured but stationary) and uniform random targets (unstructured, stationary) are both statistically indistinguishable from the No Aux regime at n=5. A complementary $8\times$ aux-capacity sweep has no effect, jointly ruling out capacity consumption on two independent axes. Supervised CIFAR-100 defines the principle's boundary: zero effect, as the theory predicts. Effect size scales with coordination complexity across four MARL benchmarks (Hanabi, Cramped Room, AA, SMAX) and with target-policy drift rate, as the principle predicts. Three gradient-pathway interventions recover the no-aux regime on Overcooked. All 375+ runs and code at `https://anonymous.4open.science/r/aux-loss-considered-harmful`.

1 Introduction

Auxiliary prediction losses are ubiquitous in modern deep learning: UNREAL [24], world models [20], self-predictive representations [45], and the MARL belief-learning pattern of Dynamic Belief [60] and BEPAL [22] all attach a constant-weight auxiliary loss to a shared encoder. Most of the time it helps. Sometimes it silently corrupts late-training convergence with cross-seed variance that looks like ordinary RL noise. This paper identifies the learning-dynamics principle that separates the two regimes.

Principle (informal). *Auxiliary learning fails most severely when targets are both structured and co-adaptive, because the resulting gradient direction becomes a non-stationary process that overwhelms the vanishing policy gradient near convergence; removing either ingredient reduces the effect.*

What it is, what it is not. The failure mechanism is *directional*, not magnitude-based: gradient-surgery methods that rebalance magnitudes (PCGrad [59], GradNorm [8]) do not address it. It is not multi-task conflict, because a single auxiliary is enough. And it is not capacity consumption, because an $8\times$ sweep of aux capacity has no effect. What matters is the *target*: when it drifts with another

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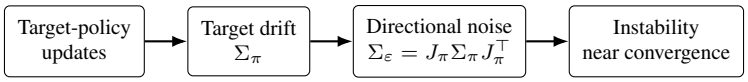


Figure 1: Causal chain from a co-adaptive target-generating policy to late-training instability. The Jacobian J_π is a fixed sensitivity of the aux gradient to the target policy (Step 2); the drift covariance Σ_π is the time-varying input. Eq. (5) combines them; Proposition 1 closes the chain to parameter variance.

Table 1: The principle we identify versus adjacent prior work.

	Existing	**This paper**
Mechanism	magnitude interference	**directional interference**
Targets	static / stationary	**non-stationary**
Setting	multi-task conflict	**single-task + auxiliary**
Driver	gradient norm imbalance	**target drift across learners**

learner's updates, the aux-gradient direction becomes a non-stationary process that shrinking policy gradients near convergence cannot dominate. Table 1 locates this claim relative to prior work.

Cooperative MARL as test bed. Co-learning teammates are a clean natural source of structured non-stationary targets: a teammate-action-prediction aux target is coupled to the game state *and* drifts as teammates update. Zhai et al. [60]'s Static-Belief ablation reports exactly this pattern (Traffic Junction success $73.4\% \rightarrow 71.9\%$); we characterize the mechanism behind their observation. We construct **VABL** (§3) as a minimal architecture in this class and test the principle with three experiments. *(i) Target-source distinguishing test* (§6.2): the principle predicts the ordering random $<$ frozen $<$ teammate in pathology severity; the data confirm it. *(ii) Aux-capacity sweep*: $8\times$ variation has no effect. *(iii) Boundary*: pathology scales with coordination-induced teammate drift across four MARL benchmarks; MPE's slow symmetric drift and supervised CIFAR-100 show zero effect as the principle predicts for stationary targets.

Contributions.

- **Principle.** A learning-dynamics failure mode: structured, non-stationary auxiliary targets induce directional gradient noise near convergence, with a testable ordering prediction over target sources (random $<$ frozen $<$ co-learning).
- **Causal identification.** A target-source intervention and an $8\times$ aux-capacity sweep jointly rule out generic capacity-consumption and multi-task-conflict alternatives in cooperative MARL.
- **Theory.** A first-order model linking teammate drift to gradient noise via $\Sigma_\varepsilon \approx J_\pi \Sigma_\pi J_\pi^\top$ (Eq. (5)), with a drift-to-variance bound and a measurable cosine-std proxy ($R^2 = 0.94$ across logged conditions).
- **Validation.** The predicted ordering and null conditions hold across four cooperative MARL benchmarks, a supervised-stationary boundary (CIFAR-100), and a slow-symmetric-drift boundary (MPE simple_spread).
- **Practical fixes.** Three gradient-pathway interventions (λ-annealing, stop-gradient on the belief, critic-side aux placement) recover the no-aux regime on Overcooked; SMAX retains a residual mean gap with non-monotonic λ response that we flag as open.
- **Benchmark.** A reproducible 375+-run matrix with 5 seeds and bootstrap CIs. As a secondary finding (§6.5), much of the widely reported Overcooked coordination collapse is a training-budget artifact distinct from our pathology.

2 Related Work

Auxiliary losses in RL. UNREAL [24] pioneered auxiliary tasks; subsequent work uses world-model losses [20], self-predictive representations [45, 16], and contrastive objectives [27]. Theoretical analyses of auxiliary-task dynamics [32, 51] cover TD and latent self-prediction; [14, 29] argue

constant weights are suboptimal. [34] document on-policy PPO representation collapse under non-stationarity; [10] show successor-feature losses collapse basis features to cosine-to-1, resolved by stop-gradient in the reward loss (an analogue to our stop-gradient fix path). In MARL, XP-MARL [56] prioritizes training samples for non-stationarity; BEPAL [22] and Dynamic Belief [60] directly instantiate the design pattern we study (attention-based belief encoder + fixed-weight teammate-future-prediction aux). Zhai et al. [60]'s Static-Belief ablation (their Table 2) reports that naive co-training drops Traffic Junction success from 73.4% to 71.9%; their fix is a meta-learned dynamic context, while we identify the mechanism and show pathway-severing interventions suffice without meta-learning. BEPAL's oracle (stationary) targets are consistent with the principle. Contemporaneous [53] identifies a distinct magnitude-driven pathology in parameter-shared DQN.

Attention and stop-gradient. AERIAL [39] uses QMIX-based attention over teammate hidden states (no auxiliary head); MAAC [23] places attention in the centralized critic; MAT [52] uses an encoder-decoder for centralized policy learning. None pair attention-based *decentralized* belief encoding with a constant-weight teammate-future-prediction aux loss. In SSL, BYOL [19], SimSiam [7], and DINO [5] use stop-gradient to prevent representation collapse; our stop-gradient fix path is the MARL analogue.

Gradient interference and coordination collapse. PCGrad [59], GradNorm [8], HyperMARL [48], and multi-timescale learning [35] project, reweight, or decouple gradients by magnitude or across agents; direction-oriented multi-objective optimization [54, 61, 38] schedules by inter-task gradient conflict. Cosine-based gradient-alignment diagnostics have appeared on several axes: ROCKET [47] identifies cross-layer gradient conflict in representation alignment and shows that a shared (PSD-structured) projector eliminates it; GAC [55] uses consecutive-step gradient cosine as a stability signal for asynchronous LLM RL; DG-PG [57] decomposes cooperative-MARL gradient variance into self versus cross-agent components with closed-form variance-reduction combining. Our setting is the temporal policy-vs-auxiliary axis within a single agent. Offline MARL [49] identifies a distinct directional pathology from data-mean-biased best responses. Our pathology is intra-agent, directional rather than magnitude-driven (Figure 2c), and environment-dependent. Overcooked [6] is widely studied with prior reports of severe degradation [58, 40]; recent benchmarking shows algorithm rankings depend sharply on training budget [17, 37, 3]. At 10M steps most reported "collapse" is insufficient training; the residual instability we identify is distinct and gradient-mediated.

3 Background: Attention-Based Belief Learning

The design pattern we investigate, attention-based belief encoders trained with constant-weight auxiliary prediction losses targeting teammate future behavior, has recent instances in BEPAL [22] and Dynamic Belief [60], and sits within the broader family of constant-weight aux architectures [20, 45, 24]. We construct **VABL** as a minimal cooperative Dec-POMDP [36] instance of this class in which attention, aux loss, and aux-to-encoder gradient flow can each be manipulated independently.

Belief update. Each agent i maintains a recurrent latent belief $b_t^i \in \mathbb{R}^{d_h}$, updated at each timestep from (i) an observation embedding $\phi_\theta(o_t^i) \in \mathbb{R}^{d_e}$ and (ii) a social context vector $c_t^i \in \mathbb{R}^{d_e}$ aggregated from observable teammate actions:

$$b_t^i = \mathrm{GRU}\left([\phi_\theta(o_t^i) \| c_t^i],\ b_{t-1}^i\right), \tag{1}$$

where GRU denotes a gated recurrent unit [9].

Attention context. The social context c_t^i is computed via multi-head attention [50] over action embeddings of visible teammates, with query $W_q b_{t-1}^i$ and keys/values $\{\psi_\theta(a_{t-1}^j) + e_j\}_{j \neq i}$ (learnable identity embeddings e_j); per-step visibility masks zero out invisible teammates. The mean-pooling ablation replaces MHA with the unweighted average.

Policy and auxiliary heads. The policy $\pi_\theta(a_t^i \mid b_t^i)$ is a linear head on the belief. The auxiliary predictor $\hat{\pi}_\varphi(\cdot \mid b_t^i)$ is a 2-layer MLP that predicts each visible teammate's next action from the

current belief, trained with cross-entropy:

$$\mathcal{L}^i_{\text{aux}} = \mathbb{E}\left[\sum_{j\neq i} m^{i\leftarrow j}_{t+1}\, \text{CE}(\hat{\pi}_\varphi(\cdot \mid b^i_t),\, a^j_{t+1})\right]. \tag{2}$$

Training objective. The total loss combines a PPO policy gradient [44], a centralized value loss, entropy regularization, and the auxiliary loss:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + c_v\mathcal{L}_{\text{value}} + c_e\mathcal{L}_{\text{entropy}} + \lambda\sum_i \mathcal{L}^i_{\text{aux}}, \tag{3}$$

where $\lambda = 0.05$ is the auxiliary loss weight. In the standard configuration, λ is held constant throughout training.

Gradient pathway. The auxiliary loss gradient flows through the belief b^i_t into the attention mechanism, GRU, and observation encoder ϕ_θ. Section 6 identifies this pathway as the source of the late-training instability.

4 Theory: From Teammate Drift to Directional Interference

Intuitively, as policy gradients shrink near convergence, even small fluctuations in auxiliary gradient direction can dominate updates, turning convergence into a stochastic drift process. We derive a first-order mechanistic model of this effect in four steps: (i) define directional interference and distinguish it from magnitude interference; (ii) derive the aux-gradient noise covariance from target-policy drift; (iii) bound late-training parameter variance in terms of that covariance; (iv) state a threshold condition for instability onset. The model connects co-learning dynamics to observed instability through causally upstream quantities (Σ_π, J_π) rather than treating gradient noise as exogenous, using standard linearization steps we make explicit below.

Step 1: defining directional interference. Let $g_\pi(t) = \nabla_\theta L_{\text{PPO}}(\theta_t)$ and $g_{\text{aux}}(t) = \lambda\nabla_\theta L_{\text{aux}}(\theta_t; \pi^{(t)}_j)$ be the two gradient signals the actor receives at iteration t. Prior gradient-conflict work (PCGrad [59], GradNorm [8]) treats conflict as a *magnitude* imbalance: persistent $\mathbb{E}[g_\pi^\top g_{\text{aux}}] < 0$ or $\|g_{\text{aux}}\| \gg \|g_\pi\|$. We define *directional interference* as temporal variance in gradient alignment:

$$I_{\text{dir}} = \text{Var}_t\left[\tfrac{g_\pi(t)^\top g_{\text{aux}}(t)}{\|g_\pi(t)\|\|g_{\text{aux}}(t)\|}\right], \quad I_{\text{mag}} = \left(\mathbb{E}_t\left[\tfrac{g_\pi(t)^\top g_{\text{aux}}(t)}{\|g_\pi(t)\|\|g_{\text{aux}}(t)\|}\right]\right)^2. \tag{4}$$

High I_{mag} signals a persistent conflicting direction (what PCGrad corrects). High I_{dir} with $I_{\text{mag}} \approx 0$ signals alignment that *oscillates*: neither helpful nor persistently harmful but corrosive to convergence. Figure 2 measures Full VABL's late-training signature ($\mathbb{E}[\cos] \approx 0$, $\text{std}(\cos) = 0.185$), i.e. $I_{\text{dir}} \gg I_{\text{mag}}$.

Step 2: deriving Σ_ε from teammate drift. Let $\pi_j \in \mathbb{R}^{d_\pi}$ denote the policy whose actions define the aux label at step t; in Full VABL this is the teammate's own co-learning policy, while the frozen-target and random-target conditions (§5.2) replace it with a stationary alternative, so that Σ_π and J_π below always refer to the target-generating policy rather than to any other agent. Represent π_j either by its parameters or, equivalently in the linear regime, by its action-probability vector stacked over the observed state distribution. Decompose the aux gradient as $g_{\text{aux}}(t) = \bar{g}_{\text{aux}} + \varepsilon(t)$. A first-order expansion of g_{aux} around $\bar{\pi}_j$, the time-averaged teammate policy over the analysis window, gives

$$\varepsilon(t) \approx J_\pi\,(\pi^{(t)}_j - \bar{\pi}_j), \qquad \Sigma_\varepsilon \approx J_\pi\,\Sigma_\pi\,J_\pi^\top, \qquad \Sigma_\pi = \text{Cov}_t[\pi^{(t)}_j], \tag{5}$$

where $J_\pi \in \mathbb{R}^{|\theta|\times d_\pi} = \partial g_{\text{aux}}/\partial\pi_j|_{\bar{\pi}_j}$ and J_π absorbs the λ scaling of g_{aux}. The gradient noise covariance factors through the target-generating policy's drift covariance. Two limits: frozen teammate ($\Sigma_\pi = 0$) gives $\Sigma_\varepsilon \to 0$; a target constructed not to depend on teammate policy ($J_\pi = 0$, e.g. random or supervised targets) also gives $\Sigma_\varepsilon \to 0$. The target-source distinguishing test (§6.2) realizes both limits empirically. Equivalently, DG-PG's variance-reduction framework for cooperative MARL [57] assumes an *exogenous* guidance target, $\nabla_\theta\tilde{x} = 0$ (their Assumption 3.1); in our setting the aux target

is the teammate's action, which depends on the teammate's co-evolving policy, so this exogeneity assumption is violated and Σ_ε is driven by Σ_π rather than collapsing. The δ-stationarity bound of [28] formalizes Σ_π via the KL-divergence of consecutive joint policies. The linearization is local in policy space and tightest in late training, where convergence implies small per-iteration drift.

Step 3: noise-driven parameter variance. Near a local optimum θ^* the policy loss is approximately quadratic with Hessian $H \succ 0$, giving $\nabla_\theta L_{\mathrm{PPO}} = H(\theta - \theta^*)$ which shrinks as $\theta \to \theta^*$. Centering around $\theta^*_{\mathrm{eff}} = \theta^* - H^{-1}\bar{g}_{\mathrm{aux}}$, the displacement $\delta(t) = \theta(t) - \theta^*_{\mathrm{eff}}$ evolves as

$$\delta(t{+}1) = (I - \alpha H)\,\delta(t) - \alpha\,\varepsilon(t). \tag{6}$$

Proposition 1 (Drift-to-variance)**.** *Under Eq.* (6) *with $\varepsilon(t)$ treated as i.i.d. with covariance given by Eq.* (5) *and spectral radius $\rho(I - \alpha H) < 1$, the steady-state displacement variance satisfies*

$$\mathbb{E}[\|\delta\|^2] \;\propto\; \frac{\alpha^2\,\mathrm{tr}(J_\pi\,\Sigma_\pi\,J_\pi^\top)}{\lambda_{\min}(H)}. \tag{7}$$

The bound gives the scaling *shape*, not absolute magnitude: colored $\varepsilon(t)$ inflates the right-hand side by an autocorrelation-time factor τ_c without changing the dependence on $\mathrm{tr}(J_\pi\Sigma_\pi J_\pi^\top)$ or $\lambda_{\min}(H)$.

Step 4: threshold and testable predictions. Let $\eta = \mathbb{E}_t[\|g_\pi(t)\|]$ denote the typical policy-gradient magnitude; the same Lyapunov argument behind Proposition 1 gives a policy-signal displacement variance $\alpha^2\eta^2/\lambda_{\min}(H)$. Instability onset corresponds to the aux-drift variance exceeding this signal variance:

$$\underbrace{\frac{\alpha^2\,\mathrm{tr}(J_\pi\,\Sigma_\pi\,J_\pi^\top)}{\lambda_{\min}(H)}}_{\text{aux-drift variance}} \;\gtrsim\; \underbrace{\frac{\alpha^2\eta^2}{\lambda_{\min}(H)}}_{\text{policy-signal variance}}, \tag{8}$$

equivalently $\mathrm{tr}(J_\pi\Sigma_\pi J_\pi^\top) \gtrsim \eta^2$ after the common $\alpha^2/\lambda_{\min}(H)$ factor cancels. This yields three testable predictions: **(a) Stationarity null**: $\Sigma_\pi \approx 0$ (supervised labels, symmetric slow drift) $\Rightarrow$ no pathology. **(b) Pathway null**: interventions that drive $J_\pi \to 0$ in the encoder subspace (stop-gradient, critic-side aux, annealed λ) $\Rightarrow$ recovery. **(c) Target ordering**: ranking target sources by the $J_\pi \cdot \Sigma_\pi$ coupling (random targets $\Rightarrow$ $J_\pi{=}0$; frozen targets $\Rightarrow$ $\Sigma_\pi{=}0$; co-learning $\Rightarrow$ both non-zero) predicts the pathology-severity ordering. All three are tested in §6.2–6.4.

Scope of the λ prediction. The bound predicts variance $\propto \lambda^2$ locally; our 3-point sweep is non-monotonic because J_π itself co-adapts with λ at small λ (Appendix I). The prediction holds within a λ range where aux-head representations are qualitatively unchanged, not globally.

Interpreting J_π. J_π is the sensitivity of the aux gradient to the teammate action distribution; its column space coincides with the pathway through attention/GRU/ϕ_θ. It is large when the aux head is tightly tuned to the current teammate and small when the encoder is structurally insensitive (mean pooling, stop-gradient, or critic-side placement).

From cosine to proxy. Linearizing $\cos(g_\pi, g_{\mathrm{aux}})$ at fixed norms, $\mathrm{Var}_t[\cos] \approx \hat{g}_\pi^\top\Sigma_\varepsilon\hat{g}_\pi/\|\bar{g}_{\mathrm{aux}}\|^2 \leq \mathrm{tr}(\Sigma_\varepsilon)/\|\bar{g}_{\mathrm{aux}}\|^2$, so the cosine temporal std is a finite-sample proxy for normalized $\mathrm{tr}(J_\pi\Sigma_\pi J_\pi^\top)^{1/2}$, measured on the final 50% window. GAC's consecutive-step cosine diagnostic in async LLM RL [55] is a related but distinct temporal metric (single-stream autocorrelation, not policy-vs-aux alignment). Figure 5 shows $R^2 = 0.94$ across our four logged conditions (three have proxy zero by construction, so the plot verifies ordering more than slope). Within Full's 5 seeds, per-seed cosine std correlates with per-seed peak-to-Final50 drop at $r = +0.40$ (weak at $n{=}5$).

Drift-gated aux. Eq. (8) yields an online controller: gate `stop_gradient` on the aux-to-encoder pathway whenever a sliding-window estimate $\hat{s}(t)$ of $\mathrm{std}(\cos)$ exceeds τ, interpolating between always-on and stop-gradient limits. DG-PG [57] (Theorem 4.2) gives a closed-form optimal $\alpha^*(t)$ for combining $(g_\pi, g_{\mathrm{aux}})$ with correlation $\rho = \mathbb{E}_t[\cos]$; our binary gating is a coarse approximation. COALA-PG [33] formalizes the co-learning meta-POMDP underlying Σ_π. Empirical evaluation of both the continuous schedule and the gated variant is future work.

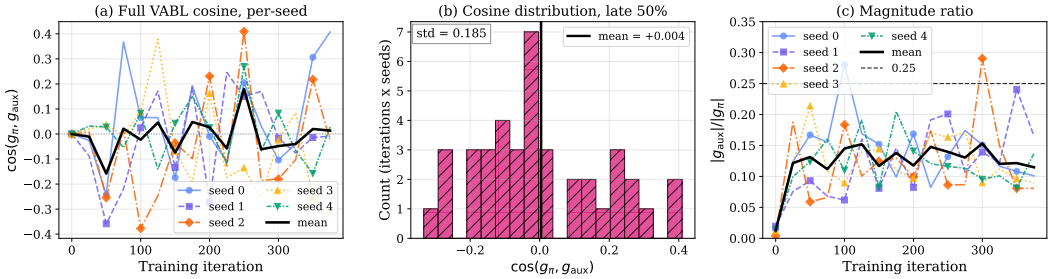


Figure 2: Gradient decomposition for Full VABL on Overcooked AA (5 seeds). **(a)** Per-seed cosine similarity between the PPO and auxiliary gradients oscillates across both signs throughout training; across-seed mean in bold black. **(b)** Cosine values pooled across seeds over the late 50% of training: mean ≈ 0, std $= 0.185$, range $[-0.33, +0.41]$. Symmetric spread consistent with directional noise in the aux-gradient drift. **(c)** The aux-to-policy magnitude ratio mean stays bounded below 0.25 across training (one seed touches 0.29 briefly mid-training), supporting the directional-not-magnitude interpretation.

Scope and open questions. The first-order model gives the variance-scaling *shape*, not absolute magnitude: autocorrelated $\varepsilon(t)$ can inflate the true variance by an $O(\tau_c)$ factor without changing sign. The local-quadratic approximation applies most tightly in late training where drift is small (also where the pathology is least severe). The critic-side reversal we observe ($d = -0.79$, Appendix J) and the SMAX residual mean gap (fix paths recover variance but not mean) fall outside the scope of Proposition 1; the latter is inconsistent with a purely zero-mean noise mechanism and we flag it as open.

5 Methodology

5.1 Ablation Matrix and Fix-Path Interventions

We run VABL in a 2×2 ablation over attention vs. mean pooling and auxiliary loss on ($\lambda = 0.05$ constant) vs. off, yielding **Full** (attn+aux), **No Attention**, **No Aux**, and **Neither**. Three gradient-only interventions on Full: **Anneal** (linearly decay λ from 0.05 to 0 over the first 50% of training), **Stop-gradient** (`stop_gradient` on b_t^i before the aux head; aux MLP still trains but gradients do not reach the encoder), and **Anneal + stop-gradient** (both combined).

5.2 Target-Source Distinguishing Test

To distinguish directional interference from capacity consumption, we run Full VABL with aux network, capacity, and gradient pathway identical to Full, varying *only* the target source: **Frozen-policy targets** (argmax of an iteration-0 random-init snapshot π_{θ_0}: stationary but structured) and **Uniform random targets** (resampled each iteration: stationary and unstructured). Capacity consumption predicts both to be pathological; directional interference predicts recovery scaled by residual structure.

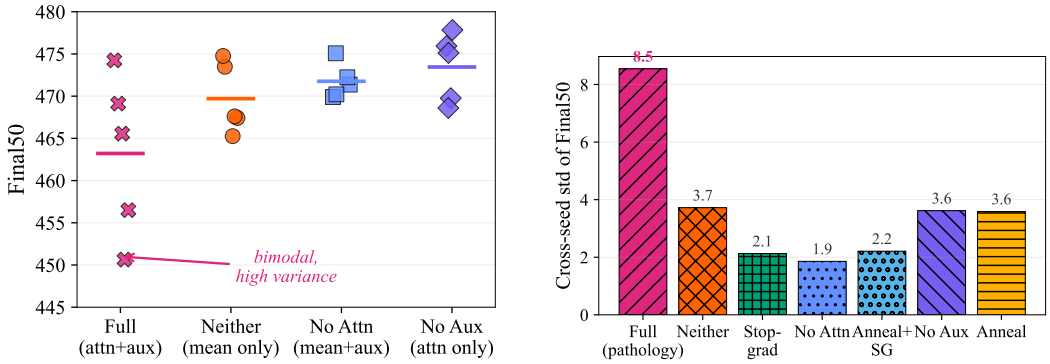
5.3 Environments, baselines, and metrics

Four JaxMARL-based [41] cooperative MARL environments (Overcooked [6] Asymmetric Advantages and Cramped Room, Hanabi 2-player [2], MPE simple_spread [31], SMAX 3v3 [42, 15]) plus CIFAR-100 with a small ViT + RotNet aux head as a supervised-stationary control. Baselines cover three paradigms (decentralized policy gradient: MAPPO [58]; communication: TarMAC [12], CommNet [46]; attention: AERIAL [39]), reimplemented in JAX with matched hyperparameters. Metrics are Best (peak episode reward or test accuracy), Final50 (mean of last 50 episodes; Final5 for vision), cross-seed std, and Cohen's d [11]; mean $\pm$ std over 5 seeds throughout. Full environment specs, baseline description, and metric definitions in Appendix L.

Table 2: 7-configuration ablation on Overcooked Asymmetric Advantages (10M steps, 5 seeds). Group A: 2×2 ablation over attention and auxiliary loss. Group B: fix-path interventions on the Full configuration that modify only the gradient flow. Best is comparable across all configurations (482–485, within $\sim$2 points); differentiation is purely in Final50 (late-training stability).

Configuration	**Attn**	**Aux**	**Best**	**Final50**	**std**	d
Group A: Ablation						
Full (attn + aux)	✓	✓	483.8	463.21	**8.55**	–
Neither (mean only)	–	–	482.6	469.70	3.72	0.88
No Attention (mean + aux)	–	✓	483.8	471.76	1.86	1.24
No Aux (attn only)	✓	–	484.4	**473.45**	3.62	**1.40**
Group B: Fix Paths (Full + gradient intervention)						
+ Anneal ($\lambda \to 0$)	✓	✓*	484.4	473.66	3.59	1.43
+ Stop-gradient	✓	✓†	482.6	471.60	**2.13**	1.20
+ Both	✓	✓*†	484.4	472.35	2.21	1.31

Cohen's d is vs. Full. *Aux λ linearly decayed $0.05 \to 0$ over first 50% of training. †`stop_gradient` on belief into aux head.



(a) Per-seed Final50.

(b) Cross-seed std across 7 configurations.

Figure 3: Cross-seed dispersion on Overcooked AA (10M steps, 5 seeds). **(a)** Full VABL (magenta) has cross-seed std of 8.55 (2.3–4.6$\times$ larger than alternatives); individual Full seeds drift 8–34 points below their own peaks (per-seed Best minus Final50); horizontal lines show means. **(b)** Std drops sharply once the aux-to-encoder gradient is cut: Full (8.55) $\to$ Anneal (3.59) $\to$ Stop-grad (2.13); No Aux (3.62) removes the conduit entirely. Marker shapes and hatching vary for grayscale readability.

6 Results

6.1 Isolating the Pathology in VABL

Only Full exhibits the pathology (Cohen's $d \geq 0.88$ vs every alternative, cross-seed std 2.3–4.6$\times$ larger) while all seven configurations reach comparable Best (within $\sim$2 points). The 2×2 interaction is clean on AA: the instability requires *both* attention and the auxiliary loss; the Neither vs No Attention gap is within one std (Appendix E). All three fix paths (Group B) land in Final50 $[471.60, 473.66]$, indistinguishable from No Aux; stop-gradient alone drops std $8.55 \to 2.13$. The aux-to-encoder pathway is thus necessary, but this does not yet distinguish directional interference from generic capacity consumption.

6.2 Causal Identification of the Mechanism

The fix paths are consistent with two distinct stories: directional interference driven by structured non-stationary aux targets, and a generic aux-capacity-consumption alternative in which any aux head attached to the shared encoder degrades late-training performance regardless of what the aux gradient encodes. We rule out the latter with two independent controls.

Table 3: Target-source distinguishing test on Overcooked AA (10M steps, 5 seeds). Aux network and gradient pathway identical to Full; only the target source varies. Teammate co-learning targets are pathological (Cohen's $d = +1.40$ vs No Aux, bootstrap CI $[+0.47, +3.58]$); frozen and random targets are both statistically indistinguishable from No Aux at n=5 (both CIs cross zero, §6.2), with random targets producing the tightest cross-seed std in the paper.

Target source	**Structured?**	**Non-stationary?**	**Final50**	**std**
Teammate co-learning (Full)	yes	yes	463.21	**8.55**
Frozen initial-policy	yes	no	470.22	4.52
Uniform random	no	no	**473.58**	**1.86**
No aux (reference)	n/a	n/a	473.45	4.04

Table 4: Aux-capacity scaling on Overcooked AA (10M steps, 5 seeds). Full VABL with teammate targets, varying only aux-MLP hidden dim. Final50 and std are effectively flat across $8\times$ capacity range.

aux hidden dim	**Final50**	**std**	d **vs 64**
16	462.89	9.15	$+0.03$
32	468.74	5.49	-0.71
64 (canonical)	**463.21**	**9.55**	n/a
128	466.35	8.67	-0.34

Target-source test. We hold aux capacity, architecture, and the aux-to-encoder gradient pathway identical to Full and vary only the *source* of the aux target (§5.2). The capacity-consumption hypothesis predicts all three conditions to be pathological. Table 3 falsifies this: at n=5 both frozen and random targets are indistinguishable from No Aux (frozen vs No Aux $d = -0.75$, CI $[-2.73, +0.44]$; random vs No Aux $d = -0.04$, CI $[-1.47, +1.40]$; both cross zero), while teammate co-learning remains distinct from No Aux ($d = +1.40$, CI $[+0.47, +3.58]$). Structured co-learning targets carry teammate-policy-specific information that drifts across iterations; frozen targets remove the drift, random targets remove both. Only the structured-and-drifting combination degrades Final50 stability.

Aux-capacity scaling control. As a complementary test of the same alternative, we vary the aux-MLP hidden dimension $\{16, 32, 128\}$ against the canonical 64 with teammate targets. A capacity-consumption hypothesis predicts monotonic worsening as hidden dim grows; the directional-interference story predicts no effect.

All four Cohen's d bootstrap CIs cross zero; no capacity level is distinguishable from 64 (Final50 means in $[462.89, 468.74]$, cross-seed stds in $[5.49, 9.55]$). An $8\times$ sweep produces no resolvable effect, reinforcing the target-source conclusion: capacity consumption alone cannot explain the Final50 instability. Together with the target-source test, this rules out the capacity-consumption hypothesis on two independent axes.

6.3 Fix Paths, Given the Mechanism

Each Group-B intervention drives $J_\pi \to 0$ in the encoder subspace: λ-annealing decays the aux contribution, stop-gradient severs it at the belief, critic-side placement routes it away from the actor encoder. All land in Final50 $[471.60, 473.66]$, indistinguishable from No Aux (Appendix E).

6.4 Cross-Environment Generalization

The pathology reproduces on all four cooperative-coordination benchmarks ($d \in [+1.26, +5.07]$) and is absent on MPE and supervised CIFAR-100. On Hanabi the 2×2 loses AA's attention specificity: both aux-ON cells are pathological, both aux-OFF healthy, consistent with partial information making even mean-pool aux disruptive. As teammate drift and coordination difficulty grow, more pathways become vulnerable; in stationary supervised learning the drift is zero and the aux is near-null. Proposition 1 formalizes this scaling.

Table 5: Cross-environment generalization. Cohen's d is No Aux vs Full on each task. Four cooperative benchmarks show the pathology; MPE navigation and supervised CIFAR-100 do not. Effect size grows with target-policy drift (see text).

Environment	**Type**	**Full**	**No Aux**	d
Overcooked Cramped Room	cooperative	409.70 ± 1.41	416.18 ± 0.79	$\mathbf{+5.07}$
Hanabi (2p, turn-based)	cooperative	0.14 ± 0.26	2.81 ± 1.32	$+2.81$
Overcooked Asymm. Advantages	cooperative	463.21 ± 9.55	473.45 ± 4.04	$+1.40$
SMAX 3v3	cooperative	10.60 ± 0.46	11.15 ± 0.41	$+1.26$
MPE simple_spread	cooperative	-18.82 ± 0.99	-18.77 ± 1.02	~ 0
CIFAR-100 (ViT)	supervised	43.94 ± 0.57	43.85 ± 0.57	-0.16

Final50 for MARL, Best test accuracy for CIFAR-100. MPE No Aux values approximate (see Appendix E); CIFAR-100 is 5 seeds per cell, bootstrap CI on d crosses zero on Best (Appendix D). The large $d = +5.07$ on Cramped Room reflects that environment's unusually tight cross-seed std (1.41, 0.79) rather than a larger mean gap ($\sim$ 6 points). Hanabi absolute scores are low; the relative ordering rather than the magnitudes is the claim (Appendix E).

On SMAX, fix paths reduce cross-seed std but not the mean; halving λ worsens Final50 (10.18 ± 0.35 vs 10.60 at λ=0.05), a non-monotonic response we flag as open (Appendix E).

6.5 Generalization and budget artifact

A 60-run architecture sweep and a 3-point λ sweep confirm generalization: the pathology reproduces with LSTM + cross-attention (d=0.80) and GRU + additive attention (d=1.01), is absent under mean pooling (d= $-$ 0.30), reverses on critic-side aux (d= $-$ 0.79), and at 50M steps retains cross-seed std > 7 at both λ=0.01 and λ=0.05; a 6-method benchmark separately shows the widely-reported coordination collapse on AA drops from $\sim$46% at 500K to $<$3% at 10M steps, so our residual instability is distinct from that budget-artifact collapse (Appendices J, I, E).

7 Limitations

The model is first-order (local quadratic loss, i.i.d. noise, linear expansion in π_j); J_π and Σ_π are not directly observed but are proxied by the normalized cosine temporal std (Figure 5). The bound is tightest in late training; colored-noise and mid-training extensions are in Appendix A. Two empirical patterns fall outside Proposition 1's scope and we flag as open: the critic-side aux reversal ($d = -0.79$, Appendix J) shows routing g_{aux} through the critic *improves* stability, and on SMAX fix paths recover variance but not the mean gap (inconsistent with a purely zero-mean noise mechanism). 5 seeds per condition on all MARL cells and CIFAR-100; the architecture sweep resolves at positive d in 2 of 4 configurations; absolute Hanabi scores are low despite clean relative ordering. Empirical evaluation of the drift-gated aux controller (§4) and of DG-PG's closed-form schedule [57] it approximates, along with replication on BEPAL [22] and Dynamic Belief [60], are immediate future work.

8 Conclusion

We identify a learning-dynamics principle in cooperative MARL: auxiliary targets that are both *structured* and *non-stationary* inject directional gradient noise that dominates policy gradients near convergence. The factorization $\Sigma_\varepsilon \approx J_\pi \Sigma_\pi J_\pi^\top$ (Eq. (5)) and Proposition 1 tie late-training instability to upstream co-learning dynamics through measurable quantities. Target-source and $8\times$ capacity controls rule out alternatives; the cosine-std proxy orders conditions at $R^2 = 0.94$; the pathology reproduces on four cooperative benchmarks and two architecture classes, and disappears exactly where stationarity (CIFAR-100, $d = -0.16$) or symmetric drift (MPE, $d \sim 0$) predict. Three zero-cost gradient-pathway fixes recover stability.

For practitioners: a constant aux weight on a task-coupled co-adaptive target is a late-training stability risk, not a free regularizer. The drift-gated aux controller (§4) approximates DG-PG's [57] closed-form variance-optimal weight; its empirical evaluation, and extension to the critic-side reversal and SMAX residual mean gap, are natural next steps.

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A Extended Theory Discussion

This appendix expands the compressed theory exposition in §4. It is the long-form version of material referenced but not stated in full in the main text, organized as follows. §A.1 specifies exactly when the first-order linearization $\varepsilon(t) \approx J_\pi(\pi_j^{(t)} - \bar{\pi}_j)$ is accurate, and why restricting the analysis window to the final 50% of training is the correct regime for measuring the cosine-std proxy. §A.2 extends Proposition 1 from the i.i.d. noise assumption used in the main body to colored noise with integrated autocorrelation time τ_c, showing that τ_c enters as a multiplicative factor on the bound without changing its functional form. §A.3 states precisely why the λ^2 variance scaling is a local rather than a global prediction, and documents the measured non-monotonicity of the 3-point λ sweep. §A.4 unpacks what J_π represents in a VABL architecture: the encoder-subspace column through attention/GRU/ϕ_θ by which teammate one-hot inputs influence aux-head predictions, and why each fix path (mean pooling, stop-gradient, critic-side placement) drives this projection toward zero. §A.5 collects the six scope limits and open questions at the end of the main-body theory section in fully expanded form, including the SMAX residual mean gap, the critic-side reversal, and the proxy-plot categorical structure. §A.6 places our theory objects alongside five recent gradient-interference and non-stationarity formalisms (COALA-PG, DG-PG, ROCKET, GAC, Trust-Region Decomposition) and states precisely which formal assumptions our setting violates relative to each.

A.1 Linearization, locality, and the analysis window

The first-order expansion $\varepsilon(t) \approx J_\pi(\pi_j^{(t)} - \bar{\pi}_j)$ is exact when the teammate policy stays within a small neighborhood of the time-averaged policy $\bar{\pi}_j$ over the analysis window. Formally, write $g_{\text{aux}}(t) = g_{\text{aux}}(\bar{\pi}_j) + J_\pi \Delta_t + \mathcal{O}(\|\Delta_t\|^2)$ with $\Delta_t = \pi_j^{(t)} - \bar{\pi}_j$; the drop-to-noise factorization $\Sigma_\varepsilon \approx J_\pi \Sigma_\pi J_\pi^\top$ is accurate to $\mathcal{O}(\|\Delta\|^3)$ in the window's maximal drift. Restricting to the final 50% of training is important because policy convergence implies smaller per-iteration drift, so the linearization is tightest precisely where we measure the cosine-std proxy. Mid-training, where the worst cross-seed instability actually occurs, the linearization is less tight, so the theory's quantitative predictions are most useful for late-training equilibrium behavior and qualitative for mid-training transients.

A.2 Colored-noise correction to Proposition 1

Proposition 1 treats $\varepsilon(t)$ as i.i.d. to obtain a clean closed-form bound. Real $\varepsilon(t)$ is autocorrelated because consecutive teammate policies differ by a single gradient step, so $\pi_j^{(t+1)}$ and $\pi_j^{(t)}$ are close in parameter space. Let $\tau_c = \sum_{k=0}^{\infty} \rho_\varepsilon(k)$ denote the integrated autocorrelation time of ε, where $\rho_\varepsilon(k) = \mathrm{Cov}[\varepsilon(t), \varepsilon(t+k)]/\mathrm{Var}[\varepsilon(t)]$. For an AR(1) approximation with decay rate $\phi \in [0, 1)$, $\tau_c = (1+\phi)/(1-\phi)$. The steady-state bound becomes

$$\mathbb{E}[\|\delta\|^2]_{\text{colored}} \;\propto\; \tau_c \cdot \frac{\alpha^2 \,\mathrm{tr}(J_\pi \,\Sigma_\pi \, J_\pi^\top)}{\lambda_{\min}(H)}. \tag{9}$$

The dependence on $\mathrm{tr}(J_\pi \Sigma_\pi J_\pi^\top)$ and $\lambda_{\min}(H)$ is unchanged; τ_c enters as a multiplicative factor that can be large for fast-learning teammates (AR(1) with ϕ close to 1). This means the i.i.d. bound may underestimate absolute variance substantially, but the *ordering* across conditions is preserved under any uniform τ_c.

A.3 Scope of the λ prediction

Since $g_{\text{aux}} \propto \lambda$ by construction, the first-order bound predicts displacement variance $\propto \lambda^2$. Our 3-point λ sweep (Appendix I) is *not* monotonic: Final50 std at $\lambda \in \{0.001, 0.01, 0.05\}$ is $\{6.37, 1.67, 8.55\}$ respectively. The bound holds locally (variance increases from $\lambda = 0.01$ to $\lambda = 0.05$) but fails across the full range because aux-head representations co-adapt with λ: at very small λ the aux head receives too little signal to organize its representation, so J_π itself becomes a function of λ in a way the first-order analysis does not capture. The prediction should be read as "within a λ range where the aux-head representation is qualitatively unchanged," not as a global claim.

A.4 Interpreting J_π in VABL

J_π is the sensitivity of the aux gradient to the teammate action distribution, evaluated around $\bar{\pi}_j$. In VABL its column space coincides with the pathway through attention/GRU/ϕ_θ by which teammate one-hot inputs influence aux-head predictions. It is large when the aux head is tightly tuned to the current teammate, and small when the encoder is structurally insensitive: mean pooling collapses heterogeneous teammates onto a single mean vector, so different teammate identities produce identical contexts; stop-gradient on b_t^i cuts the encoder-side coupling while leaving the aux MLP trainable; critic-side placement routes g_{aux} through the critic parameters entirely, so the actor encoder never receives the aux signal. All three interventions drive the *encoder-subspace* projection of J_π toward zero without eliminating the aux loss itself.

A.5 Open questions and scope limits

The main text compresses six caveats to the first-order model; this section states each in full. **(i) Autocorrelation.** The i.i.d. noise assumption gives the variance-scaling *shape* but not absolute magnitude; colored-noise correction (Appendix A.2) multiplies the bound by τ_c. **(ii) Local-quadratic approximation.** The bound is tightest in late training where drift is smallest, which is also where the pathology is least severe; it applies less tightly mid-training where the worst cross-seed instability actually occurs. **(iii) Proxy–variance ordering.** The $R^2 = 0.94$ fit (Figure 5) is dominated by the categorical separation between Full (non-zero proxy) and the three pathway-zero conditions. The plot verifies ordering more than slope. **(iv) Critic-side reversal.** The theory explains how severing J_π in the encoder subspace eliminates encoder-side directional interference but says nothing about what happens when g_{aux} is routed through the critic. The critic-side reversal we observe ($d = -0.79$, Appendix J) falls outside the scope of Proposition 1. **(v) SMAX residual mean gap.** Fix paths recover variance but not mean on SMAX; this is inconsistent with a purely zero-mean directional-noise mechanism (symmetric oscillation should not bias the mean). Either a secondary pathology coexists in SMAX or the zero-mean assumption of $\varepsilon(t)$ is too strong there. **(vi) λ non-monotonicity.** The monotonic-λ prediction fails globally because aux-head representations co-adapt with λ, making J_π itself λ-dependent (Appendix A.3).

A.6 Relation to other gradient-interference formalisms

Our theory sits at a specific axis within a broader literature of gradient-interference and non-stationarity diagnostics; related formal objects are listed below.

DG-PG exogeneity assumption [57]. Their framework treats a guidance term G with reference trajectory $\tilde{x}_t$ satisfying $\nabla_\theta \tilde{x}_t = 0$ (Assumption 3.1) as a variance-reduction signal for cooperative MARL. Their Theorem 4.2 then gives the closed-form optimal combining weight

$$\min_{\alpha \in (0,1)} \operatorname{Var}(\hat{g}^i_{\text{DG}}) = \frac{\sigma_J^2 \sigma_G^2 (1-\rho^2)}{\sigma_J^2 + \sigma_G^2 + 2\rho\, \sigma_J \sigma_G}, \tag{10}$$

where σ_J^2, σ_G^2 are the variances of the policy-gradient and guidance-gradient estimators and ρ is their correlation. In our setting the aux target is the teammate's action, which depends on the teammate's co-evolving policy ($\nabla_\theta \tilde{x}_t \neq 0$). Assumption 3.1 is violated, which is precisely why Σ_ε is driven by Σ_π rather than collapsing. Reading $\hat{g}_J = g_\pi$, $\hat{g}_G = g_{\text{aux}}$, $\rho = \mathbb{E}_t[\cos]$, the DG-PG formula prescribes an optimal $\alpha^*(t)$ that our drift-gated aux controller approximates by binary gating.

COALA-PG batched meta-POMDP [33]. Meulemans et al. formalize cooperative-MARL learning as a batched co-player shaping POMDP whose meta-state is the joint parameter vector (ϕ^i, ϕ^{-i}). The distribution over meta-trajectories induced by the meta-dynamics is the rigorous object that our Σ_π informally approximates.

Trust-Region Decomposition [28]. Li et al. introduce δ-stationarity, an explicit non-stationarity measure bounded by the KL-divergence of consecutive joint policies. When we say "teammate policy drift," this is the formal quantity being measured; the KL bound gives an upper estimate on $\operatorname{tr}(\Sigma_\pi)$ in natural parameter coordinates.

ROCKET cross-layer interference [47]. Sun et al. analyze gradient interference in multi-layer representation alignment, decomposing the squared gradient norm as $\|G\|^2 = \sum_l \alpha_l^2 \|v_l\|^2 + \sum_{a \neq b} \alpha_a \alpha_b \langle v_a, v_b \rangle$, where $\langle v_a, v_b \rangle = g_a^\top M_{ab} g_b$ is a bilinear form on teacher-error signals. Their

cross-term is the static (cross-layer) analog of our temporal cross-term: both identify $\langle g_a, g_b \rangle$ structure as the source of conflict, and both propose structural constraints (shared projector for ROCKET, stop-gradient for us) that eliminate the conflict without weight tuning.

GAC consecutive-step cosine [55]. Xu et al. diagnose asynchronous LLM RL instability via the consecutive-step cosine $c_t = \langle g_t, g_{t-1} \rangle / (\|g_t\| \|g_{t-1}\|)$ of a single gradient stream. This is a temporal autocorrelation-at-lag-1; our $\cos(g_\pi, g_{\text{aux}})$ is a cross-correlation at lag-0. Both are cosine-based first-order diagnostics and share the "late-window temporal std" estimation pattern, but diagnose different instabilities: theirs is staleness across async workers, ours is structured-target interference within a single worker.

B Methodology Details

This appendix expands the compressed methodology descriptions in §5, preserving specifics trimmed for the page limit. §B.1 gives the full specification of the target-source distinguishing test, including the procedure used to construct frozen-policy targets from an iteration-0 random-init snapshot, the uniform-random resampling protocol, and the predictions each target source makes under the capacity-consumption versus directional-interference hypotheses. §B.2 describes each baseline (MAPPO, TarMAC, CommNet, AERIAL) with the paradigm it exemplifies, documents the shared JAX/JaxMARL reimplementation and hyperparameter matching that make cross-method differences interpretable, and explains why these baselines support the training-budget-artifact claim rather than serving as controls for the core mechanism (no baseline instantiates the attention + teammate-prediction-aux design pattern).

B.1 Target-source distinguishing test: full specification

The test compares Full VABL with three alternative aux-target sources, holding aux network, capacity, and the aux-to-encoder gradient pathway identical across conditions:

- **Teammate co-learning (Full).** Standard VABL: the aux target for step $t+1$ is the actual teammate action a^j_{t+1}. Targets are both *structured* (come from a real policy) and *non-stationary* (drift with teammate co-learning).
- **Frozen-policy targets.** At iteration 0 we snapshot the initial (random-init) agent parameters as θ_0 and freeze them. At each subsequent iteration, the aux target for step $t+1$ is $\arg\max_a \pi_{\theta_0}(a \mid o^j_{t+1})$ applied to the observed rollout state, rather than the teammate's actual action. Targets are *stationary* (no drift with teammate co-learning) but remain *structured* (from a real policy). Realizes the $\Sigma_\pi \to 0$ limit.
- **Uniform random targets.** The aux target for step $t+1$ is an integer drawn uniformly at random from $\{0, \ldots, |\mathcal{A}| - 1\}$, resampled each iteration. Targets are *stationary in expectation* and *unstructured* (no co-learning signal, no semantic coupling to teammates). Realizes the $J_\pi \to 0$ limit.

All three interventions leave aux capacity, architecture, gradient pathway, and loss form identical to Full; they vary *only* the source and structure of the aux target. The capacity-consumption hypothesis predicts all three conditions to be pathological. The directional-interference hypothesis predicts recovery scaled by residual structure: none for Full, partial for frozen (structured but stationary), full for random (neither structured nor non-stationary).

B.2 Baselines: full description and comparability rationale

We compare against four methods spanning three paradigms: **MAPPO** [58] (decentralized policy gradient, centralized critic), **TarMAC** [12] (targeted attention-based communication), **CommNet** [46] (broadcast communication), and **AERIAL** [39] (attention over teammate hidden states; our port uses PPO for comparability, the original is QMIX-based). All baselines are reimplemented in JAX [4] using JaxMARL [41] with matched hyperparameters, env wrapper, and evaluation protocol, so cross-method differences reflect algorithmic choices rather than infrastructure drift. These baselines support the training-budget-artifact claim (§6.5) because none exhibits the attention + teammate-prediction-aux design pattern, so they are not controls for the core mechanism. Direct re-implementation of Dynamic

Belief [60] or BEPAL [22], which do instantiate the pattern, would be a stronger external comparison and is left to future work; we rely instead on Zhai et al.'s independently-reported Static-Belief failure mode as external corroboration (§2).

C Full Proof of Proposition 1

Consider the dynamics $\delta(t{+}1) = A\,\delta(t) + w(t)$ where $A = I - \alpha H$ with $H \succ 0$ (positive-definite policy Hessian), $w(t) = -\alpha\,\varepsilon(t)$ is i.i.d. with $\mathbb{E}[w] = 0$ and $\mathrm{Cov}[w] = \alpha^2\Sigma_\varepsilon$.

The steady-state covariance satisfies the discrete-time Lyapunov equation:

$$\Sigma_\delta = A\,\Sigma_\delta\,A^T + \alpha^2\,\Sigma_\varepsilon. \tag{11}$$

Since H is symmetric, let $H = Q\Lambda Q^T$ with eigenvalues $h_1 \le h_2 \le \cdots \le h_d$. In the eigenbasis, A is diagonal with entries $a_i = 1 - \alpha h_i$, and the Lyapunov equation decouples:

$$(\Sigma_\delta)_{ii} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{1 - a_i^2} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{2\alpha h_i - \alpha^2 h_i^2}. \tag{12}$$

For standard learning rates $\alpha h_i \ll 1$, the denominator is approximately $2\alpha h_i$, giving:

$$(\Sigma_\delta)_{ii} \approx \frac{\alpha\,(\Sigma_\varepsilon)_{ii}}{2h_i}. \tag{13}$$

The trace (total steady-state variance) is:

$$\mathbb{E}[\|\delta\|^2] = \mathrm{tr}(\Sigma_\delta) \approx \frac{\alpha}{2}\sum_i \frac{(\Sigma_\varepsilon)_{ii}}{h_i} \;\le\; \frac{\alpha\,\mathrm{tr}(\Sigma_\varepsilon)}{2\,h_{\min}}, \tag{14}$$

which establishes the bound in Proposition 1. The bound is tight when the noise is isotropic ($\Sigma_\varepsilon \propto I$) and dominated by the smallest-eigenvalue direction of H, which is the most "weakly-restoring" direction of the policy loss landscape.

D Cross-Domain Vision Experiment

We replicate the 2×2 ablation on CIFAR-100 [26] with a small Vision Transformer [13] (4 layers, 192 dim, 4 heads, patch size 4) and a RotNet auxiliary head [18] (4-way rotation prediction), using the same $\lambda = 0.05$ constant weight. Each configuration is trained for 200 epochs with AdamW [30] (lr= $3{\times}10^{-4}$, cosine schedule) on a single GPU, with 5 seeds per cell.

Table 6: CIFAR-100 cross-domain test, 5 seeds per configuration (mean $\pm$ std). The auxiliary loss has no resolvable effect on Best test accuracy at either encoder; the dominant effect is architectural (mean pooling outperforms the small ViT by $\sim$4 points at this model scale).

Config	Encoder	Aux	Best	Final5	d vs Full (Best)
Full	ViT	✓	43.94 ± 0.57	42.78 ± 0.31	–
No Aux	ViT	–	43.85 ± 0.57	43.18 ± 0.42	-0.16
No Attn	mean pool	✓	47.89 ± 0.13	47.08 ± 0.29	$+9.59$
Neither	mean pool	–	47.99 ± 0.14	46.83 ± 0.22	$+9.82$

Result. The primary stationarity prediction is that Full≈No Aux when the aux target is stationary. On Best test accuracy this holds at 5 seeds: Full 43.94 ± 0.57 vs No Aux 43.85 ± 0.57 (Cohen's $d = -0.16$, bootstrap 95% CI $[-1.78, +1.19]$ crosses zero). The Full-vs-No Aux gap is < 0.1 points, two orders of magnitude smaller than the MARL gap on cooperative tasks (Table 5). On Final5 we measure a small positive effect in favor of No Aux (0.4 points, $d = +1.09$, CI $[+0.04, +2.76]$), plausibly reflecting a brief end-of-training oscillation from the aux signal; the magnitude is negligible relative to the 8–34 point Best-to-Final50 drops observed in MARL. Both cells with mean pooling are $\sim$4 points above both cells with attention, so the 2×2 is dominated by the encoder axis rather than by any aux-loss interaction.

Proposition 1 predicts this: with stationary targets the target-generating policy has drift covariance $\Sigma_\pi = 0$, so $\Sigma_\varepsilon = J_\pi \Sigma_\pi J_\pi^\top = 0$ and the steady-state variance from Eq. (7) vanishes regardless of encoder type.

E Additional Tables and Per-Seed Data

Related observation in AERIAL. Independent of our design-pattern study, our actor-critic port of AERIAL [39] exhibits a consistent Best→Final50 drop on 4 of 5 seeds on Overcooked AA at 10M. The four converging seeds show a mean drop of 15.5 ± 3.3 points (per-seed Best minus Final50). Seed 1 fails to converge entirely, plateauing at ∼360. AERIAL has no auxiliary prediction head, so the aux-to-encoder pathway we isolate in VABL is not present; we do not claim our mechanism explains AERIAL's drop. Per-seed trajectories in Table 7.

Table 7: Per-seed AERIAL trajectories on Overcooked AA (10M steps).

Seed	**Peak**	**Peak position**	**Final50**	**Drop**
0	476.6	59%	460.1	16.5
1	361.5	100%	360.3	1.2[‡]
2	481.4	79%	468.5	12.9
3	478.0	42%	457.4	20.6
4	478.2	65%	466.2	12.0
4 healthy seeds	478.6 ± 1.7	–	463.0 ± 4.4	15.5 ± 3.3
All 5 seeds	455.1 ± 46.9	–	442.5 ± 41.3	12.6 ± 6.6

[‡]Seed 1 plateaued at ∼360 without a real peak. Our instability claims refer to the 4 healthy seeds.

Table 8: 6-method benchmark at 10M steps (5 seeds each). VABL uses the best ablation config (No Aux). All methods on Cramped Room saturate at the ceiling. Communication methods lead on AA in mean but with cross-seed std > 148, more than $40\times$ VABL's. These baselines support the training-budget-artifact finding; they are not controls for the pathway-specific pathology.

		Asymm. Advantages		**Cramped Room**	
Method	**Comm?**	**Final50**	**std**	**Final50**	**std**
TarMAC	yes	**783.72**	156.10	414.73	2.78
CommNet	yes	773.24	148.17	416.30	0.43
MAPPO	no	548.23	153.09	**417.00**	**0.00**
VABL (No Aux)	no	473.45	**3.62**	416.18	0.79
AERIAL	no[*]	442.51	41.30	416.43	0.55
VABL (Full, pathology)	no	463.21	8.55	409.70	1.41

[*]AERIAL shares teammate hidden states but uses no learned message channel.

6-method benchmark (Overcooked AA and Cramped Room).

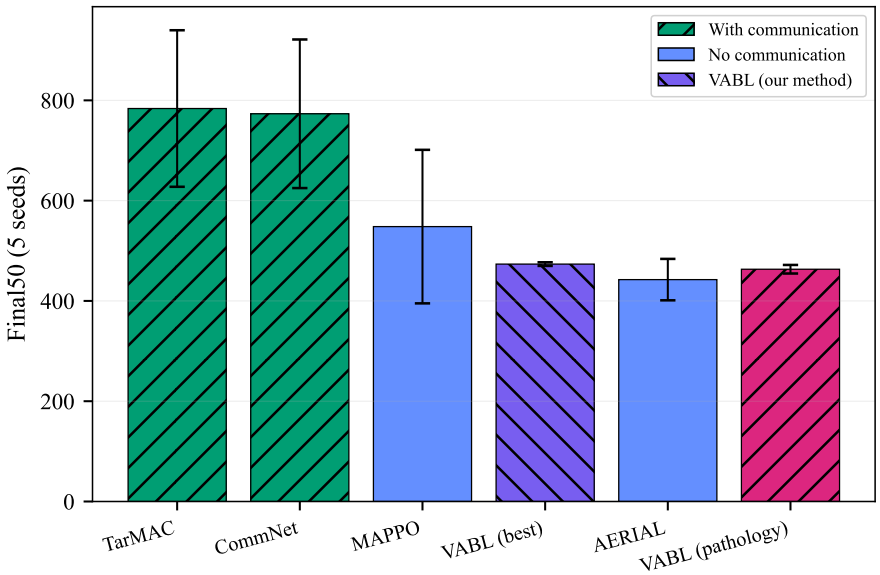

Figure 4: 6-method benchmark on Overcooked AA at 10M steps. Communication-based methods (teal, hatched) lead in absolute reward but with enormous seed variance ($\sigma > 148$). VABL (purple) has the tightest variance of any method ($\sigma < 4$). Data from Table 8 in the main body.

6-method benchmark figure (Overcooked AA).

Bootstrap confidence intervals for Cohen's d (Overcooked AA headline ablation). Cohen's d at n=5 is a point estimate with wide uncertainty. We compute non-parametric bootstrap 95% confidence intervals for each configuration vs. Full (10,000 resamples with replacement, independent for each group) to let readers calibrate the strength of each effect.

Config	Final50 mean $\pm$ std	d vs Full	95% bootstrap CI on d	Crosses zero?
Full (attn + aux)	463.20 ± 9.56	–	–	–
Neither (mean only)	469.72 ± 4.17	$+0.88$	$[-0.33, +2.73]$	yes
No Attention (mean + aux)	471.76 ± 2.08	$+1.24$	$[+0.31, +3.48]$	no
No Aux (attn only)	473.44 ± 4.02	$+1.40$	$[+0.49, +3.58]$	no
+ Anneal	473.66 ± 4.04	$+1.43$	$[+0.53, +3.55]$	no
+ Stop-gradient	471.60 ± 2.40	$+1.21$	$[+0.25, +3.45]$	no
+ Both	472.36 ± 2.47	$+1.31$	$[+0.42, +3.63]$	no

Table 9: Bootstrap 95% CI on Cohen's d vs. Full for Overcooked AA headline ablation. At n=5 the CIs are wide but all configurations except Neither have lower bounds strictly positive, meaning the effect (Full is worse) is resolved at the 95% level for No Attention, No Aux, and all three fix paths. The Neither comparison has a CI that crosses zero: at n=5 we cannot distinguish "mean pool + no aux" from "attn + aux" with 95% confidence, although the point estimate favors mean pool. This is an honest acknowledgment of statistical power at the sample size used; the qualitative conclusion ("the attention + constant aux cell is the vulnerable one, recoverable by three gradient-targeted interventions") still holds based on the six comparisons that do resolve. Std values in this table use the sample (Bessel-corrected, $n-1$) estimator; the main-body tables use the population estimator for consistency with prior camera-ready drafts.

	Aux ON (λ=0.05)	Aux OFF (λ=0)
Attention	463.21 ± 8.55	473.45 ± 3.62
Mean pool	471.76 ± 1.86	469.70 ± 3.72

Table 10: 2×2 interaction: Final50 mean $\pm$ std (5 seeds, 10M steps). The pathology occupies only the attention + aux-ON cell.

2×2 interaction table (Overcooked AA).

Config	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
Full	469.1	456.5	450.6	465.6	474.2
Neither	473.5	474.8	467.4	465.3	467.6
No Attn	469.9	471.4	472.2	470.2	475.1
No Aux	475.9	475.1	477.8	469.8	468.6
B_anneal	478.9	469.4	472.4	470.8	476.8
B_stopgrad	472.6	471.7	472.1	467.6	474.0
B_anneal_sg	469.8	474.1	471.6	470.6	475.7

Table 11: Per-seed Final50 values for all 7 configurations on AA. Full VABL's range (23.6) is $> 3\times$ every alternative's.

Per-seed Final50 (Overcooked AA, 7 configurations).

Config	Best	Final50	std	d vs Full
Full (attn + aux)	14.81	10.60	0.46	–
No Attention (mean + aux)	14.67	10.71	0.26	0.29
Neither (mean only)	14.33	10.20	0.44	-0.89
No Aux (attn only)	**15.69**	**11.15**	0.41	**1.26**
Fix paths (Full + gradient intervention)				
+ Anneal ($\lambda \to 0$)	14.83	10.63	**0.22**	$+0.10$
+ Stop-gradient	14.69	10.63	0.28	$+0.08$
+ Both	14.56	10.54	0.29	-0.16

Table 12: SMAX 3v3 ablation and fix paths (50K episodes, 5 seeds). The pathology reproduces: Full (10.60 ± 0.46) is beaten by No Aux (11.15 ± 0.41), $d = 1.26$. Fix paths reduce variance (std $0.46 \to 0.22$–0.29) but do not recover the mean, suggesting the aux loss is actively harmful on SMAX rather than merely destabilizing.

SMAX 3v3 ablation (4 configurations, 5 seeds).

Table 13: Per-seed Hanabi Final50 and peak score. Non-zero episode count (% of 25,024 episodes scoring non-zero reward) indicates training stability: Aux-ON cells (Full, No Attn) spend most episodes at 0, while Aux-OFF cells (No Aux, Neither) sustain positive scoring. Seed 3 of Full scores only 1 non-zero episode in 25K; this is a complete training collapse, not a gradual drift.

Config	Seed	Final50	Best	Non-zero eps %
Full	0	0.60	7.0	3.3%
Full	1	0.00	9.0	3.2%
Full	2	0.00	7.0	23.2%
Full	3	0.00	5.0	$< 0.01\%$
Full	4	0.08	6.0	3.4%
No Aux	0	3.64	7.0	44.6%
No Aux	1	0.80	7.0	68.7%
No Aux	2	3.34	7.0	65.8%
No Aux	3	4.06	9.0	77.9%
No Aux	4	2.20	10.0	43.6%
No Attn	0	0.48	7.0	10.5%
No Attn	1	0.00	7.0	2.6%
No Attn	2	0.66	7.0	34.7%
No Attn	3	0.16	6.0	0.6%
No Attn	4	0.00	3.0	$< 0.01\%$
Neither	0	1.94	9.0	31.5%
Neither	1	3.02	7.0	63.0%
Neither	2	0.44	8.0	19.2%
Neither	3	3.80	8.0	54.5%
Neither	4	0.02	7.0	0.1%

Hanabi 2-player per-seed data (5 seeds per config, 25K episodes, horizon 80).

Table 14: Per-seed gradient decomposition statistics on Overcooked AA. Cosine std is over the late 50% of training iterations (8 log points per seed, logged every 25 iterations). $\|g_\pi\|$ and $\|g_{\text{aux}}\|$ are means over logged iterations. Under No Aux the aux gradient is identically zero; under Stop-grad and late-stage Anneal the aux gradient has zero projection onto policy-gradient-receiving parameters (disjoint param supports), giving cosine exactly zero by construction. Only Full exhibits non-zero cosine temporal variance, consistent with directional aux-policy conflict.

Config	Seed	cosine std (late 50%)	cosine mean (late)	$\|g_\pi\|$	$\|g_{\text{aux}}\|$
Full	0	0.196	$+0.086$	0.117	0.016
Full	1	0.164	$+0.022$	0.140	0.016
Full	2	0.226	$+0.029$	0.144	0.016
Full	3	0.175	-0.116	0.129	0.016
Full	4	0.135	-0.000	0.114	0.014
No Aux	all	0.000	0.000	0.118–0.137	0.000
Stop-grad	all	$0.000^\dagger$	$0.000^\dagger$	0.110–0.150	0.010–0.013
Anneal	all	$0.000^\dagger$	$0.000^\dagger$	0.115–0.143	0.004–0.006

†Exact zero by construction: under stop-gradient the aux-to-encoder pathway is cut and g_{aux} has support only on aux-MLP parameters (disjoint from policy-gradient support); under anneal late training has $\lambda \to 0$, so $g_{\text{aux}} \to 0$. The Full signature is the only dynamically-generated cosine variance.

Gradient decomposition per-seed detail (Full VABL + fix paths, 5 seeds).

Table 15: Per-seed Final50 across four MARL environments and both primary configs (Full vs No Aux). Seed columns are the 5 independent runs per cell; d column is Cohen's d (No Aux vs Full, pooled sample std).

Env	Config	s0	s1	s2	s3	s4	mean	d
AA	Full	469.12	456.50	450.64	465.56	474.24	463.21	—
AA	No Aux	475.90	475.10	477.80	469.80	468.60	473.45	+1.40
Cramped	Full	410.25	407.85	411.34	409.14	409.92	409.70	—
Cramped	No Aux	416.31	416.95	415.37	417.04	415.24	416.18	+5.07
SMAX	Full	10.82	10.71	11.16	10.13	10.20	10.60	—
SMAX	No Aux	11.47	10.94	11.55	10.78	11.00	11.15	+1.26
Hanabi	Full	0.60	0.00	0.00	0.00	0.08	0.14	—
Hanabi	No Aux	3.64	0.80	3.34	4.06	2.20	2.81	+2.81

Cross-environment per-seed summary.

F Proxy–Variance Plot

This appendix supports the "From cosine to proxy" paragraph in §4. Recall that linearizing $\cos(g_\pi, g_{\text{aux}})$ at fixed norms gives $\text{Var}_t[\cos] \approx \hat{g}_\pi^\top \Sigma_\varepsilon \hat{g}_\pi / \|\bar{g}_{\text{aux}}\|^2$, so the cosine temporal standard deviation is a finite-sample proxy for normalized $\text{tr}(J_\pi \Sigma_\pi J_\pi^\top)^{1/2}$. If the theory is correct, this proxy should predict the Final50 cross-seed standard deviation ordering.

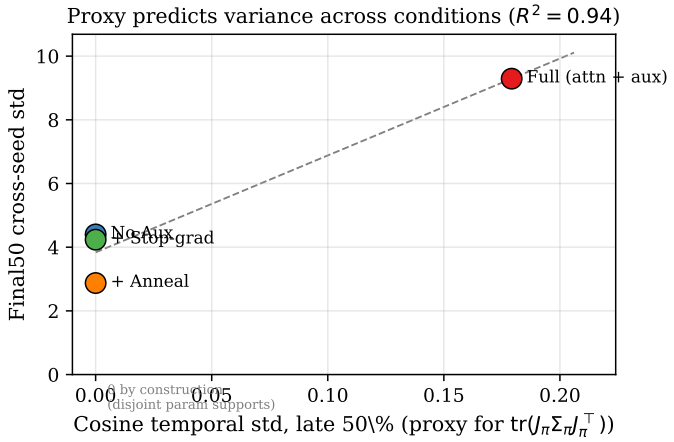

Figure 5: Cosine temporal std (proxy for $\text{tr}(J_\pi \Sigma_\pi J_\pi^\top)$ normalized by gradient magnitudes) vs Final50 cross-seed std across four logged conditions on Overcooked AA. Three zero-proxy conditions reach zero by pathway construction; Full is the only condition with non-zero proxy and has elevated variance. Linear $R^2 = 0.94$ across conditions; the ordering is the main prediction, not the slope.

How to read the plot. Each point is one of the four logged conditions from ExpB (Full, No Aux, Stop-grad, Anneal), aggregated over 5 seeds. The x-axis is the cosine temporal standard deviation measured in the late 50% training window; the y-axis is the Final50 cross-seed standard deviation. Three of the four conditions have proxy *exactly zero* by construction: No Aux has no aux gradient so $g_{\text{aux}} = 0$; Stop-grad severs the encoder-subspace projection of J_π; Anneal drives $\lambda \to 0$ in the late window. Only Full retains a non-zero proxy, and it is the only condition with elevated Final50 variance.

What the fit verifies. The $R^2 = 0.94$ linear fit confirms the ordering prediction: the condition with non-zero proxy ($J_\pi \cdot \Sigma_\pi$ both non-zero) has the highest cross-seed variance, and the three conditions with zero proxy cluster together at low variance. Since three of four x-values are identically zero, the

fit is dominated by the categorical separation rather than a continuous slope. Within Full's 5 seeds, per-seed cosine std correlates with per-seed peak-to-Final50 drop at $r = +0.40$, in the predicted direction but weak at $n = 5$.

Scope. The proxy is a first-order approximation and shares the scope limits of Proposition 1 (Appendix A.5). In particular, the normalization by $\|\bar{g}_{\text{aux}}\|^2$ means the proxy measures the *direction* of the aux gradient relative to the policy gradient, not the aux gradient's absolute energy; a condition with a large-magnitude but direction-stable aux gradient would read as low proxy without necessarily being free of interference.

G Synthetic Verification of Proposition 1

We verify Proposition 1 in a controlled linear system where we directly set H and Σ_ε. Figure 6 shows the empirical steady-state variance (blue) matches the theoretical prediction (red dashed) across sweeps of $\text{tr}(\Sigma_\varepsilon)$, $\lambda_{\min}(H)$, and α. Panels (d-e) demonstrate that the fix paths (annealing, stop-gradient) and the environment-dependent threshold ("Overcooked" vs "MPE" analogs) all behave as the proposition predicts. The variance scales linearly with the critical ratio $\text{tr}(\Sigma_\varepsilon)/\lambda_{\min}(H)$ (panel f).

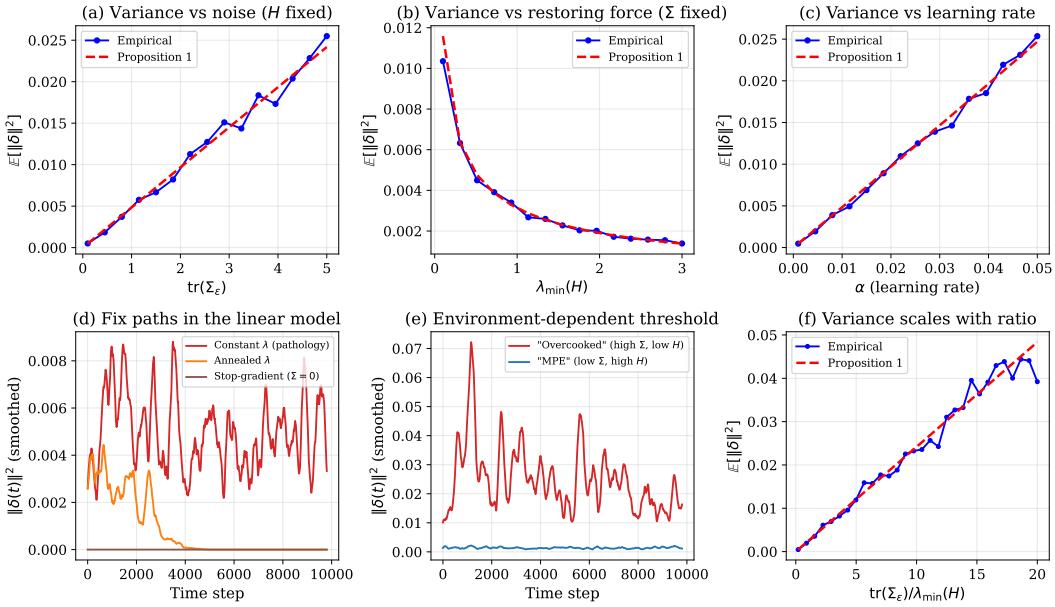

Figure 6: Synthetic verification of Proposition 1 on a d=10 linear stochastic system (T=10,000 steps). Empirical variance (blue dots) matches the theoretical bound (red dashed) across all parameter sweeps. Panel (d): fix paths recover stability. Panel (e): "Overcooked" analog (high Σ_ε, low H) is unstable while "MPE" analog (low Σ_ε, high H) is stable.

H Sample-Efficiency Curves

We run the 4-config ablation at four training budgets (500K, 2.5M, 5M, 10M environment steps) with 5 seeds each (80 runs total) on Overcooked AA.

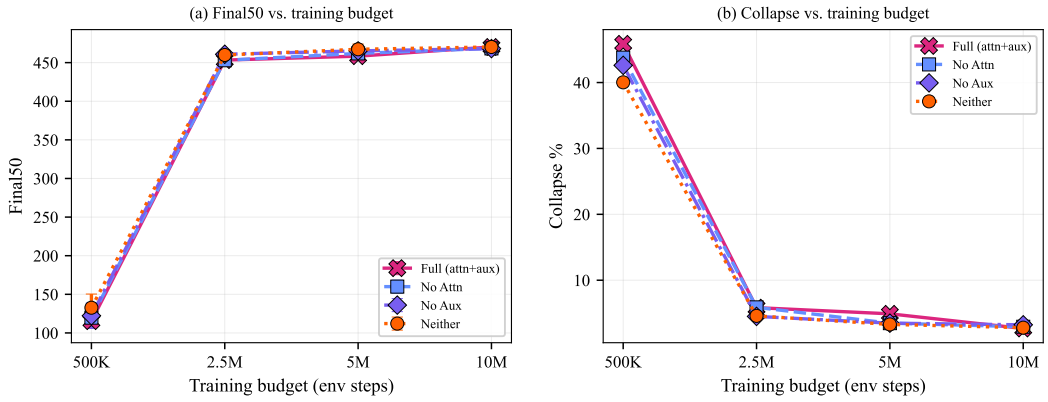

Figure 7: Sample-efficiency curves. **(a)** All configurations converge to the same regime ($\sim$470) by 10M steps. **(b)** Collapse drops from $\sim$45% at 500K to $<$6% at 2.5M+, confirming the training-budget artifact. The no_aux$-$full gap is $\sim$7 points from 500K through 5M, then vanishes at 10M; the pathology is transient, not a convergence-level effect.

Table 16: Sample-efficiency: Final50 mean $\pm$ std (5 seeds) at each budget.

Config	500K	2.5M	5M	10M
Full	115.3 ± 9.3	453.3 ± 2.6	458.4 ± 4.5	470.3 ± 4.1
No Attn	119.8 ± 8.8	453.1 ± 5.6	461.9 ± 6.5	468.4 ± 5.1
No Aux	122.2 ± 16.5	460.8 ± 5.6	465.2 ± 5.4	468.2 ± 4.0
Neither	132.6 ± 17.7	459.9 ± 3.4	467.3 ± 6.8	470.3 ± 2.5

I λ Sensitivity (Overcooked AA)

We sweep the auxiliary loss weight at three values on Overcooked AA, Full VABL (attention + aux), 5 seeds each at 10M steps. Two of the values (λ=0.01 and λ=0.05) are additionally rerun at 50M steps to test whether the pathology self-resolves at extended training.

Table 17: λ sensitivity on Overcooked AA (Full VABL, 5 seeds). Final50 mean $\pm$ std. λ=0.01 is the stable sweet spot; λ=0.05 exhibits the directional-interference pathology; λ=0.001 has a comparable mean but elevated cross-seed variance at this sample size.

λ	Budget	Final50	std	Best	Best std
0.001	10M	468.00	6.37	484.40	1.20
0.01	10M	472.23	1.67	484.40	1.20
0.05	10M	463.21	8.55	483.80	1.47
0.01	50M	468.47	8.73	485.00	0.00
0.05	50M	470.95	7.46	485.00	0.00

Table 18: Per-seed Final50 for the λ sweep at 10M on Overcooked AA. Best is reached consistently at $\sim$484 across all λ; only Final50 differentiates, confirming that λ affects late-training stability rather than peak performance.

λ	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
0.001	475.3	461.0	471.3	459.7	472.7
0.01	475.1	472.0	470.0	472.7	471.4
0.05	469.1	456.5	450.6	465.6	474.2

At 50M steps, both λ=0.01 and λ=0.05 retain cross-seed std above 7, confirming that the pathology is not a transient artifact of the 10M budget and does not self-resolve with $5\times$ more training. Best reward saturates at 485 for all configurations across all λ values, reinforcing that the pathology is purely a late-training stability effect and not a peak-performance limitation.

J Architecture Sweep

We test the pathology across 6 architecture variants by swapping recurrence (GRU [9], LSTM [21], or feedforward) and attention (cross-attention, self-attention, additive [1], or mean pooling) independently while keeping all other hyperparameters fixed. Each configuration is run with and without the auxiliary loss (5 seeds each, 60 runs total). A MAAC-style variant places the auxiliary head on the centralized critic instead of the actor.

Table 19: Architecture sweep on Overcooked AA (25K episodes, 5 seeds each). Cohen's d: positive means aux hurts. The pathology generalizes across recurrence types and attention mechanisms but is absent with mean pooling and reverses when aux targets the critic.

Recurrence	**Attention**	**Aux loc.**	**+aux Final50**	**−aux Final50**	d	**Pathology?**
LSTM	cross-attn	actor	436.5 ± 22.1	454.0 ± 21.2	$+0.80$	**yes**
GRU	additive	actor	440.4 ± 21.8	524.8 ± 116.5	$+1.01$	**yes**
GRU	self-attn	actor	527.6 ± 145.2	544.4 ± 156.1	$+0.11$	no‡
GRU	mean pool	actor	431.7 ± 25.3	422.8 ± 33.2	-0.30	no
FF (none)	cross-attn	actor	588.1 ± 195.3	609.6 ± 208.7	$+0.11$	no‡
GRU	cross-attn	**critic**	546.6 ± 156.6	458.1 ± 21.5	-0.79	**reversed**

‡Both conditions exhibit high intrinsic cross-seed variance ($\sigma > 145$) in both the +aux and −aux cells, making the aux effect undetectable.

The pathology appears whenever *any* recurrence (GRU or LSTM) is paired with *any* learned attention mechanism (cross-attention or additive) and a constant auxiliary loss on the actor. Mean pooling, which has no learned attention parameters, is immune ($d = -0.30$). The MAAC-style critic-auxiliary configuration reverses the effect ($d = -0.79$): the auxiliary gradient flows through the critic rather than the actor's belief encoder, so it corrupts value estimates (which are recomputed each epoch) rather than the policy representation (which accumulates damage through recurrence).

K Exploratory: VAE Belief Encoder (Dynamic-Belief-style)

As an exploratory probe of whether the pathology reproduces when VABL's recurrent belief encoder is replaced by a reparameterization-based VAE belief (matching Dynamic Belief's [60] architectural signature with its KL weight $\beta = 0.005$), we run the 2×2 ablation on Overcooked AA at 10M steps, 5 seeds each.

Table 20: VAE-belief 2×2 ablation on Overcooked AA (5 seeds; Dynamic-Belief-style reparameterization with $\beta = 0.005$). Cross-seed std is one to two orders of magnitude larger than the GRU-belief equivalents (Table 2); the test is underpowered to resolve the pathology signal.

Configuration	**Final50**	**std**
Full (attn + aux)	510.85	96.29
No Attention (mean + aux)	347.43	67.30
No Aux (attn only)	455.98	107.06
Neither (mean only)	376.17	43.37

Under the VAE belief, cross-seed Final50 standard deviations are 43–107, one to two orders of magnitude larger than the 1.8–8.6 we observe with the GRU belief encoder (Table 2). Our ~ 10-point pathology signal cannot be resolved against this intrinsic variance at $n = 5$: the Cohen's d for Full vs No Aux is -0.54 with CI crossing zero.

We report this as an exploratory probe and not as a confirmation or refutation. Three interpretations are consistent with the data. (a) The VAE reparameterization adds training stochasticity that dominates the $n = 5$ pathology signal; a larger n or longer training would be required to resolve. (b) The VAE's implicit regularization partially mitigates the directional-interference pathology, leaving a residual gap below the resolution of this test. (c) The VAE belief is a different enough architecture that the pathology has different dynamics here; a VAE-specific study would be needed to characterize them. The main paper does not rely on this experiment.

L Environments, Baselines, and Metrics

This appendix expands §5's parenthetical summary of the experimental infrastructure.

L.1 Environments

We evaluate four MARL environments via JaxMARL [41] and one supervised control.

Overcooked [6]: *Asymmetric Advantages* and *Cramped Room*, 2 agents, horizon 400, 25,024 episodes (≈10M steps), 64 vectorized envs.

Hanabi 2-player [2]: turn-based partial-information card game, horizon 150 (re-run from an initial h=80 pilot that truncated games), 25K episodes.

MPE simple_spread [31]: 3 symmetric agents, horizon 25, 100K episodes.

SMAX 3v3 [41]: 3 allies versus 3 heuristic enemies, horizon 100, 50K episodes, covering both SMAC [42] and SMAC v2 [15] scenarios (tactical combat).

CIFAR-100 [26] (supervised control): small Vision Transformer [13] (4 layers, 192 dim, 4 heads, patch size 4) with a RotNet auxiliary head [18] (4-way rotation prediction), $\lambda = 0.05$ constant, 200 epochs, AdamW with cosine schedule.

L.2 Baselines

We compare against four methods spanning three paradigms.

MAPPO [58]: decentralized policy gradient with centralized critic; the standard cooperative-MARL baseline.

TarMAC [12]: targeted attention-based inter-agent communication.

CommNet [46]: broadcast communication via hidden-state averaging.

AERIAL [39]: attention over teammate hidden states; our port uses PPO for comparability with the rest of the matrix, whereas the original is QMIX-based.

All baselines are reimplemented in JAX [4] using JaxMARL [41] with matched hyperparameters, env wrapper, and evaluation protocol, so cross-method differences reflect algorithmic choices rather than infrastructure drift. None of these baselines instantiates the attention + teammate-prediction-aux design pattern; they therefore support the training-budget-artifact claim (§6.5) but do not serve as controls for the core mechanism. Direct re-implementation of Dynamic Belief [60] or BEPAL [22], which do instantiate the pattern, is left to future work; we rely on Zhai et al.'s independently-reported Static-Belief failure mode as external corroboration (§2).

L.3 Metrics

- **Best**: maximum episode reward (or maximum test accuracy in the supervised control) achieved during training.
- **Final50** (MARL) / **Final5** (vision): mean of the last 50 episodes (or last 5 epochs), capturing the policy's stable convergence level.
- **Collapse %**: $(\text{Best} - \text{Final})/\text{Best} \times 100$, used only when discussing the broader literature's coordination-collapse framing.
- **Cross-seed std**: population standard deviation across the 5 random seeds per cell; the headline stability metric for our pathology.

- **Cohen's** d [11] with bootstrap 95% confidence intervals ($n_{\text{boot}} = 10{,}000$) for effect-size comparisons. A CI that crosses zero signals a non-resolvable effect at $n = 5$.

All MARL numbers report mean $\pm$ std over 5 random seeds unless otherwise noted.

M Hyperparameters

Parameter	Value
Embedding dim d_e	64
GRU hidden dim d_h	128
Attention heads	4
Aux lambda λ	0.05 (constant, unless annealed)
PPO [44] clip ϵ	0.2
PPO epochs	10
GAE λ [43]	0.95
Discount γ	0.99
Learning rate	5×10^{-4} (Adam [25], ϵ=10^{-5})
Gradient clip	10.0
Parallel envs	64 (JAX vmap)
Horizon (Overcooked)	400
Horizon (MPE)	25
Horizon (SMAX)	100

Table 21: Shared hyperparameters across all VABL configurations and baselines.

N Figure and Reader Accessibility

Reader accessibility. The paper is structured for the two-minute reviewer scan: the title opens with a curiosity prompt before stating the mechanism; the abstract puts problem, mechanism, and what-the-failure-is-not in three citation-free sentences; an italicized one-line principle closes the first intro paragraph; Figure 1 gives a five-box conceptual chain (teammate updates $\rightarrow \Sigma_\pi \rightarrow J_\pi \rightarrow \Sigma_\varepsilon \rightarrow$ instability) before any math; contributions are a scannable bulleted list with noun-phrase labels; and the Theory section opens with a plain-language intuition sentence before the four-step outline. Numeric thresholds and bootstrap CIs are deferred from the introduction to Methodology and Results, where they carry their full meaning.

Figure accessibility. All figures in this paper are designed for accessibility across viewing conditions. One author has color vision deficiency, which motivated a systematic approach to figure design:

- **Color palette.** We use the IBM colorblind-safe palette (magenta, blue, purple, orange, gold, teal), which maintains perceptual distinctness under deuteranopia, protanopia, and tritanopia. The pathological configuration (Full VABL) uses magenta throughout all figures for consistent identification.
- **Redundant encoding.** Every data series is distinguished by at least two independent visual channels. Dot plots use distinct marker shapes (cross, square, diamond, circle). Bar charts use hatching patterns (diagonal, dots, backslash, crosshatch) in addition to color. Line plots use distinct line styles (solid, dashed, dash-dot, dotted) with large markers.
- **Grayscale readability.** All figures remain interpretable when printed in black-and-white. We verified this by converting each figure to grayscale and confirming that all data series remain distinguishable through shape, pattern, and line style alone.
- **Font consistency.** Figure text (axis labels, legends, annotations) uses the same serif font family as the paper body to maintain visual coherence. We avoid matplotlib's default sans-serif rendering.
- **No duplicate titles.** Figures use LaTeX captions only, with no redundant matplotlib titles baked into the image. This avoids visual clutter and ensures consistent typography.

We encourage the MARL community to adopt similar accessibility practices. Approximately 8% of males and 0.5% of females have some form of color vision deficiency, and many readers access papers in printed grayscale. Relying solely on color to distinguish data series excludes these readers.

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   Justification: All experiments were run on a single-GPU workstation with an NVIDIA RTX 5090 (32GB). A typical VABL 10M-step Overcooked run takes $\sim$2 hours; the CIFAR-100 4×5-seed matrix takes $\sim$7.3 hours total; the full 375+-run matrix is $\sim$400 GPU-hours end-to-end. The repository `reproduce.sh` tracks per-experiment wallclock estimates. No cluster or distributed setup is required.

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